

Project TERANODE: Bill of Materials & Cost Estimation

Definitive Component & Price Checklist

S.No.	Device / Sensor / Material	Quantity	Estimated Price (NPR)
1. Processing, Logic & Telemetry Core			
1	Raspberry Pi 4 (8GB RAM Model)	1 Unit	
2	Official Raspberry Pi 4 Charger (Type-C, 5V 3A)	1 Unit	
3	Micro SD Card (64GB, High-Endurance)	1 Unit	
4	Arduino Uno R3 (For local analog pH parsing)	1 Unit	
5	ESP8266 Wi-Fi Module (NodeMCU / ESP-01)	1 Unit	
2. Industrial Soil Analytics & Liquid Array			
6	Industrial RS485 Soil NPK Sensor	1 Unit	

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7	MAX485 TTL to RS485 Converter Module	1 Unit	
8	Liquid pH Value Detection Sensor Module (BNC Probe + Amplifier)	1 Unit	
9	Corrosion-Resistant Capacitive Soil Moisture Sensor (V1.2)	1 Unit	
10	ADS1115 16-Bit 4-Channel I2C ADC Module	1 Unit	
3. Atmospheric & Weather Meteorological Array			
11	DHT22 High-Precision Digital Temperature & Humidity Sensor	1 Unit	
12	DS18B20 Waterproof Probe Temperature Sensor	1 Unit	
13	BMP280 Digital Barometric Pressure & Altitude Sensor	1 Unit	
14	Dedicated Tipping-Bucket Style	1 Unit	

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	Rain Sensor Module		
4. Enclosure & Field Infrastructure			
15	IP65 Waterproof Sealed Junction Enclosure Box	1 Unit	
16	Miscellaneous Hookup Wires, Terminal Blocks, and Hardware	1 Kit	
TOTAL PROJECTED HARDWARE BUDGET			